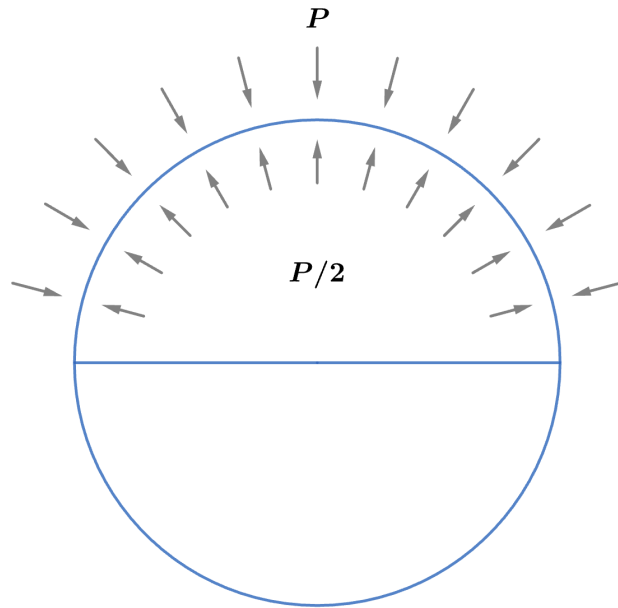
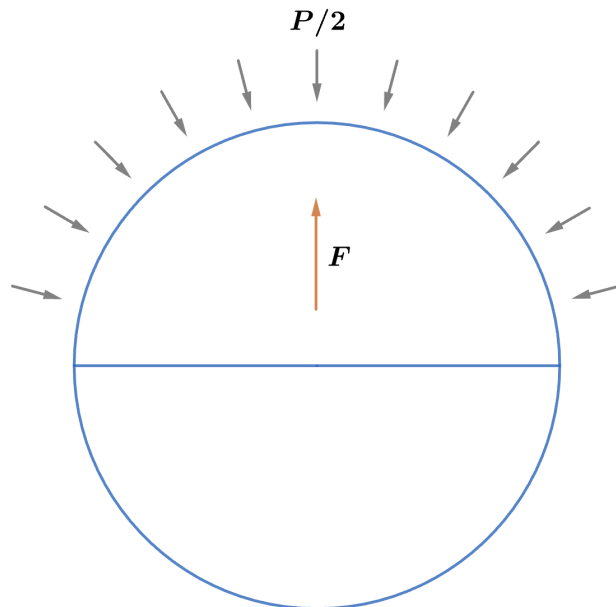


2024 F=ma Exam: Problem 11

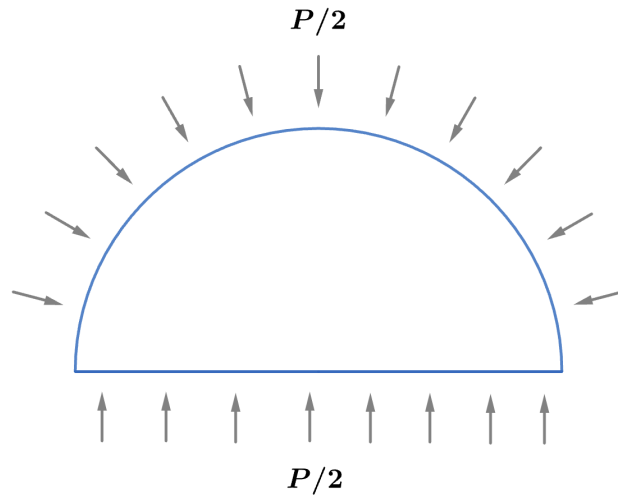
Kevin S. Huang



When the sphere is returned to sea level, the air pressure outside is P while the air pressure inside is $P/2$. At any point on the sphere, the force on the outside is double the force on the inside so it is equivalent to consider air outside to be at pressure $P/2$ and vacuum inside the sphere.



We claim the net force from the pressure on the hemispherical surface is equal to the net force from the same pressure applied to the bottom circular surface of the hemisphere. This is true because we can consider placing the hemispherical shell in a fluid of uniform pressure $P/2$ (assuming no gravity).



Since the hemisphere is at rest, the force on the bottom must balance the force from the top. In our case, this force is what's needed to separate the hemispheres. Thus,

$$F = (P/2)A = \frac{P}{2}(\pi r^2) = \frac{10^5 \text{ Pa}}{2}\pi(0.4 \text{ m})^2 = 25,000 \text{ N}$$

so the answer is A.